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CREATE DATABASE IF NOT EXISTS LibraryManagementSystem;

USE LibraryManagementSystem;

CREATE TABLE Books (
    book_id INT PRIMARY KEY,
    title VARCHAR(100) NOT NULL,
    author VARCHAR(100) NOT NULL,
    availability BOOLEAN NOT NULL DEFAULT TRUE
);

CREATE TABLE Students (
    student_id INT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    course VARCHAR(50),
    contact VARCHAR(20)
);

CREATE TABLE Librarians (
    librarian_id INT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    employee_code VARCHAR(30) UNIQUE
);

CREATE TABLE Transactions (
    transaction_id INT AUTO_INCREMENT PRIMARY KEY,
    book_id INT NOT NULL,
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    student_id INT NOT NULL,
    issue_date DATE NOT NULL,
    due_date DATE NOT NULL,
    return_date DATE DEFAULT NULL,
    FOREIGN KEY (book_id) REFERENCES Books(book_id),
    FOREIGN KEY (student_id) REFERENCES Students(student_id),
    CHECK (due_date >= issue_date),
    CHECK (return_date IS NULL OR return_date >= issue_date)
);

INSERT INTO Books (book_id, title, author, availability)
VALUES
(101, 'C++ Programming', 'Bjarne Stroustrup', TRUE),
(102, 'Database System Concepts', 'Abraham Silberschatz', TRUE),
(103, 'Operating System Concepts', 'Peter Galvin', TRUE);

INSERT INTO Students (student_id, name, course, contact)
VALUES
(1, 'Rahul Sharma', 'BCA', '9876543210'),
(2, 'Aman Singh', 'BCA', '9876543211');

INSERT INTO Librarians (librarian_id, name, employee_code)
VALUES
(1, 'Library Admin', 'LIB001');

SELECT * FROM Books;
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SELECT *
FROM Books
WHERE book_id = 101;

SELECT *
FROM Books
WHERE title LIKE '%C++%';

SELECT *
FROM Books
WHERE author LIKE '%Stroustrup%';

SELECT *
FROM Books
WHERE availability = TRUE;

SELECT * FROM Students;

START TRANSACTION;

UPDATE Books
SET availability = FALSE
WHERE book_id = 101
AND availability = TRUE;

SET @book_issued = ROW_COUNT();
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INSERT INTO Transactions
   (book_id, student_id, issue_date, due_date)
SELECT
   101,
   1,
   CURRENT_DATE,
   DATE_ADD(CURRENT_DATE, INTERVAL 14 DAY)
WHERE @book_issued = 1
AND EXISTS (
   SELECT 1
   FROM Students
   WHERE student_id = 1
);

SET @transaction_created = ROW_COUNT();

COMMIT;

SELECT
   t.transaction_id,
   b.title,
   s.name AS student_name,
   t.issue_date,
   t.due_date
FROM Transactions t
JOIN Books b ON t.book_id = b.book_id
JOIN Students s ON t.student_id = s.student_id
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WHERE t.return_date IS NULL;

START TRANSACTION;

UPDATE Transactions
SET return_date = CURRENT_DATE
WHERE book_id = 101
AND student_id = 1
AND return_date IS NULL;

SET @book_returned = ROW_COUNT();

UPDATE Books
SET availability = TRUE
WHERE book_id = 101
AND @book_returned = 1;

COMMIT;

SELECT
    t.transaction_id,
    b.title AS book_title,
    s.name AS student_name,
    t.issue_date,
    t.due_date,
    t.return_date,
    CASE
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        WHEN t.return_date IS NULL THEN 'Issued'
        ELSE 'Returned'
    END AS status
FROM Transactions t
JOIN Books b ON t.book_id = b.book_id
JOIN Students s ON t.student_id = s.student_id;

SELECT
    b.title,
    s.name AS student_name,
    t.due_date
FROM Transactions t
JOIN Books b ON t.book_id = b.book_id
JOIN Students s ON t.student_id = s.student_id
WHERE t.return_date IS NULL
AND t.due_date < CURRENT_DATE;

SELECT COUNT(*) AS total_books
FROM Books;

SELECT COUNT(*) AS total_students
FROM Students;

UPDATE Books
SET title = 'Advanced C++ Programming',
    author = 'Bjarne Stroustrup'
WHERE book_id = 101;
```

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DELETE FROM Books
WHERE book_id = 103;
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